Srinath Ranganathan

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Summary

Data professional with hands-on experience in data engineering, analytics, artificial intelligence (AI), and business intelligence within the pharmaceutical commercial domain. Skilled in building data pipelines, developing AI/ML solutions, automating workflows, and delivering reliable data solutions using Python, SQL, and Databricks. Seeking opportunities to contribute to data-driven decision-making and scalable analytics and AI systems.

Education

KGiSL Institute of Technology, Affiliated to Anna University — Coimbatore, TN

Bachelor of Technology in Artificial Intelligence and Data Science — *Nov. 2021 – May 2025*

Current CGPA: 7.8 — HSC: 87% — SSLC: 81%

Work Experience

Databricks Data Warehouse Migration

- Contributed to the end-to-end migration of a SQL Server data warehouse to Databricks, including data ingestion from SFTP files and REST APIs using PySpark and Spark SQL.
- Automated workflows using Databricks Jobs and implemented logging and validation frameworks to ensure reliable data processing.

Tableau REST API Automation

- Automated distribution of Tableau story and dashboard reports to client employees on weekly and monthly schedules using the Tableau REST API with Python.
- Developed proof-of-concept data validation processes between SQL Server databases and Tableau reports to ensure data accuracy and reporting consistency.

AI / ML Proof of Concepts

- Developed proof-of-concept models for anomaly detection and data validation using unsupervised algorithms such as Isolation Forest, One-Class SVM, and K-Means.
- Built a Python framework to extract visuals from exported BI reports and validate them using locally hosted vision and language models.

REST API Integration — Porzio Expense Reporting

- Integrated a new REST API data source for the Porzio expense reporting system, supporting bidirectional data transfer between external systems and the enterprise data warehouse.
- Built end-to-end data pipelines for data ingestion, transformation, and outbound reporting workflows.

Python and Excel Report Automation

- Developed Python scripts to execute SQL queries, generate Excel reports, and automate report delivery through email.
- Reduced manual reporting effort by 90% by standardizing automated reporting processes.

Data Support

- Retrieved and processed datasets from CMS government repositories to support pharmaceutical analytics and commercial decision-making.
- Assisted the data operations team in validating data consistency between SQL Server databases and Tableau dashboards.

Technical Skills

Primary Skills

Programming & Data Processing: Python, SQL, PySpark.
Data Analysis & Visualization: Microsoft Excel, Tableau, Power BI, Pandas, NumPy, Matplotlib, Seaborn, Plotly.
Machine Learning & Statistics: TensorFlow, Keras, Scikit-learn, Statsmodels.
Data Engineering & Workflow Tools: Airflow, MLflow, Git, GitHub.
Cloud & Platforms: AWS.

Secondary Skills

Databases: MongoDB, Cassandra.
Web Scraping & Automation: BeautifulSoup, Selenium, Regular Expressions.
Data Versioning & Experiment Tracking: DVC, DagsHub.
Other Tools: Microsoft 365.

Certifications

- Databricks Certified Data Engineer Associate — Credential Link
- IBM Data Science Professional Certificate — Credential Link
- Tableau Business Intelligence Analyst Specialization — Credential Link
- IBM Introduction to Machine Learning Specialization — Credential Link

Projects

Real-Time Clickstream Recommendation System (Final Year Project)
Technologies: Kafka, Spark, Cassandra, Flask, Power BI

- Designed and implemented a real-time clickstream data pipeline using Kafka, Spark, and Cassandra to capture and process user interaction events.
- Built a Flask-based web application to simulate e-commerce user activity and generate personalized product recommendations.
- Stored clickstream data in Cassandra and managed user and product metadata using MySQL for scalable data management.
- Generated business insights from clickstream interaction data using Power BI dashboards.

Image Search Web Application (CLIP + FAISS + EasyOCR)

- Developed an image-text search system using CLIP embeddings and FAISS for fast similarity-based retrieval.
- Integrated EasyOCR to extract text from images and enable searchable metadata indexing.
- Designed the system to support deployment on a Raspberry Pi home server for local image storage and privacy-focused data management.

Zomato Restaurant Data Analysis

- Performed data cleaning and preprocessing on a large restaurant dataset containing ratings, cuisines, and pricing information.
- Conducted exploratory data analysis (EDA) using Python and Pandas to identify trends in customer preferences and restaurant performance.
- Handled missing values, duplicates, and inconsistent records to improve dataset quality for analysis and reporting.

Elephant Detection & Security System (YOLOv8)

- Developed a real-time animal detection system using the YOLOv8 object detection model.
- Achieved approximately 94% detection accuracy in live camera-based wildlife monitoring scenarios.
- Implemented classification for four animal species: elephant, zebra, rhino, and buffalo.

Additional Information

Soft Skills
Quick Learner
Adaptability
Team Collaboration
Problem Solving

Hobbies
3D Modeling
IoT Projects